Exp10: Set up monitoring and logging for a Kubernetes cluster using Prometheus and Grafana.

Step 1: Check Prerequisites

Check kubectl

```
kubectl version --client
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ kubectl version --client
Client Version: v1.35.3
Kustomize Version: v5.7.1
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Check Minikube

```
minikube status
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

cse@cse-B760M-H:~/CT_LAB/exp10$
```

If not running the run the following command:

```
minikube start
```

Step 2: Install Helm (One-time)

```
curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ curl https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 11929  100 11929    0     0   187k      0 --:--:-- --:--:-- --:--:--  190k
Downloading https://get.helm.sh/helm-v3.20.1-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
[sudo] password for cse:
helm installed into /usr/local/bin/helm
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Check:

```
helm version
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ helm version
version.BuildInfo{Version:"v3.20.1", GitCommit:"a2369ca71c0ef633bf6e4fccd66d634eb379b371", GitTreeState:"clean", GoVersion:"go1.25.8"}
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Step 3: Add Prometheus Community Repo

```
helm repo add prometheus-community https://prometheus-
community.github.io/helm-charts
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
"prometheus-community" has been added to your repositories
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Update:

```
helm repo update
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "prometheus-community" chart repository
Update Complete. ⎈Happy Helming!⎈
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Step 4: Install Prometheus + Grafana

```
helm install monitoring prometheus-community/kube-prometheus-stack
```

This installs:

- Prometheus
- Grafana
- Alertmanager
- Node Exporter

```
cse@cse-B760M-H:~/CT_LAB/exp10$ helm install monitoring prometheus-community/kube-prometheus-stack
NAME: monitoring
LAST DEPLOYED: Tue Apr  7 15:15:54 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
kube-prometheus-stack has been installed. Check its status by running:
  kubectl --namespace default get pods -l "release=monitoring"

Get Grafana 'admin' user password by running:

  kubectl --namespace default get secrets monitoring-grafana -o jsonpath="{.data.admin-password}" | base64 -d ; echo

Access Grafana local instance:

  export POD_NAME=$(kubectl --namespace default get pod -l "app.kubernetes.io/name=grafana,app.kubernetes.io/instance=monitoring" -oname)
  kubectl --namespace default port-forward $POD_NAME 3000

Get your grafana admin user password by running:

  kubectl get secret --namespace default -l app.kubernetes.io/component=admin-secret -o jsonpath="{.items[0].data.admin-password}" | base64 --decode ; echo


Visit https://github.com/prometheus-operator/kube-prometheus for instructions on how to create & configure Alertmanager and Prometheus instances using the Operator.
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Step 4: Wait for Pods

```
kubectl get pods
```

Wait until all are **Running**

```
cse@cse-B760M-H:~/CT_LAB/exp10$ kubectl get pods
NAME                                                     READY   STATUS    RESTARTS      AGE
alertmanager-monitoring-kube-prometheus-alertmanager-0   2/2     Running   0             61s
monitoring-grafana-7b5d4d6d96-h7qgx                      3/3     Running   0             74s
monitoring-kube-prometheus-operator-5d5965bdc6-p2vd2     1/1     Running   0             74s
monitoring-kube-state-metrics-67d5f7bf68-zspqp           1/1     Running   0             74s
monitoring-prometheus-node-exporter-ppxbc                1/1     Running   0             74s
myapp-6849dcb48b-ncw2v                                   1/1     Running   1 (24h ago)   24h
myapp-6849dcb48b-thkg9                                   1/1     Running   1 (24h ago)   24h
myapp-6849dcb48b-vbvt7                                   1/1     Running   1 (24h ago)   24h
prometheus-monitoring-kube-prometheus-prometheus-0       2/2     Running   0             61s
webapp-645b5b46cd-8cpkq                                  1/1     Running   0             45m
cse@cse-B760M-H:~/CT_LAB/exp10$
```

Step 6: Check Services

```
kubectl get services
```

```
cse@cse-B760M-H:~/CT_LAB/exp10$ kubectl get services
NAME                                      TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)                      AGE
alertmanager-operated                     ClusterIP   None             <none>        9093/TCP,9094/TCP,9094/UDP   108s
kubernetes                                ClusterIP   10.96.0.1        <none>        443/TCP                      15d
monitoring-grafana                        ClusterIP   10.108.228.50    <none>        80/TCP                       2m1s
monitoring-kube-prometheus-alertmanager   ClusterIP   10.107.5.211     <none>        9093/TCP,8080/TCP            2m1s
monitoring-kube-prometheus-operator       ClusterIP   10.100.114.182   <none>        443/TCP                      2m1s
monitoring-kube-prometheus-prometheus     ClusterIP   10.102.223.82    <none>        9090/TCP,8080/TCP            2m1s
monitoring-kube-state-metrics             ClusterIP   10.104.96.93     <none>        8080/TCP                     2m1s
monitoring-prometheus-node-exporter       ClusterIP   10.96.23.144     <none>        9100/TCP                     2m1s
prometheus-operated                       ClusterIP   None             <none>        9090/TCP                     108s
webapp                                    NodePort    10.100.76.179    <none>        80:32469/TCP                 41m
cse@cse-B760M-H:~/CT_LAB/exp10$
```

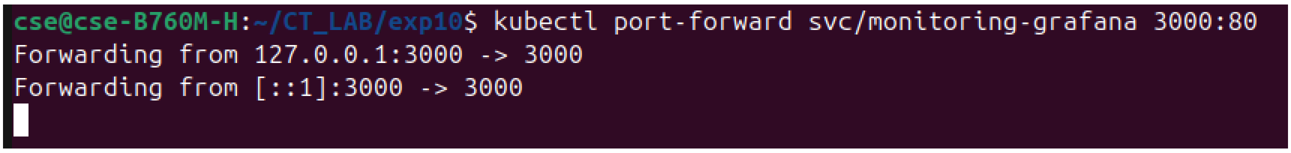
Look for:

```
monitoring-grafana
monitoring-kube-prometheus-prometheus
```

Step 7: Access Grafana

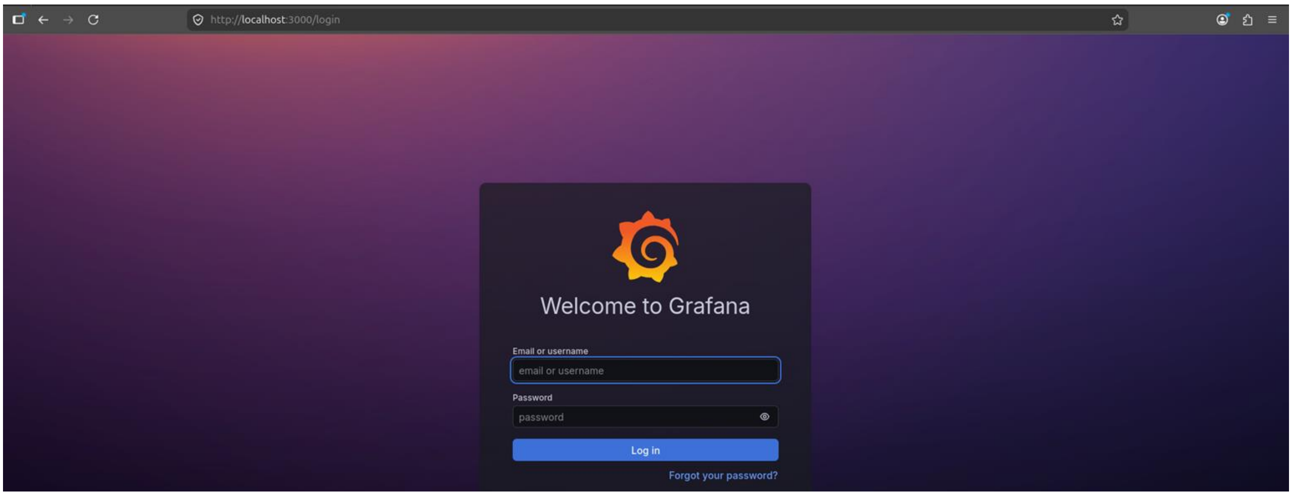
Option 1 (Recommended)

```
kubectl port-forward svc/monitoring-grafana 3000:80
```



Open browser:

```
http://localhost:3000
```



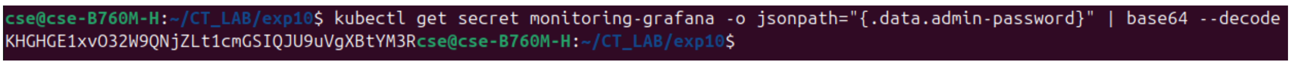
Step 8: Get Grafana Login

Username:

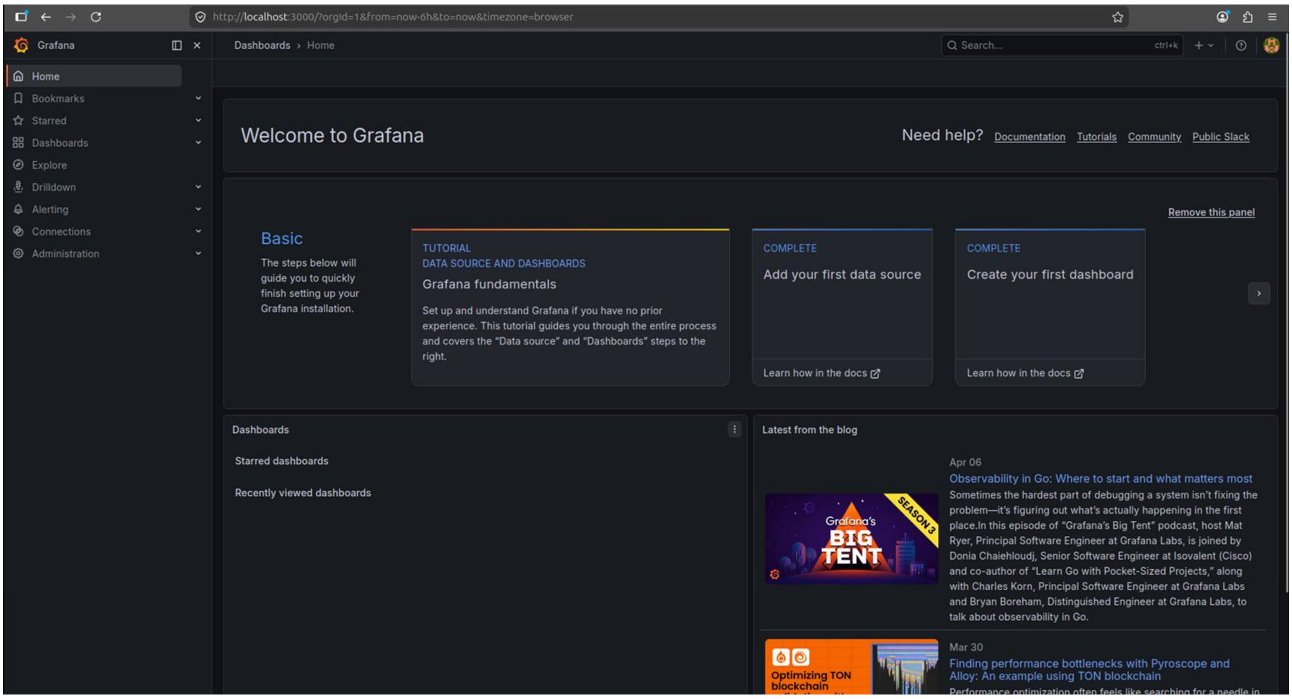
```
admin
```

Open a 2nd terminal and run the following command to get the password:

```
kubectl get secret monitoring-grafana -o jsonpath="{.data.admin-password}" | base64 --decode
```



Copy the password and paste it in login window of Grafana.



Step 9: Explore Dashboard

After login:

Go to:

Dashboards → Browse

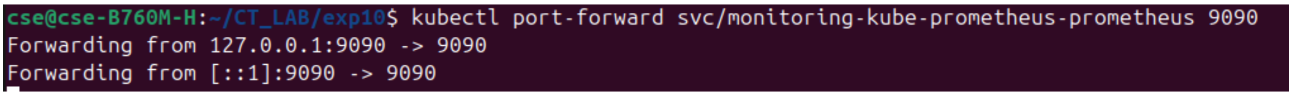
You will see:

- Kubernetes cluster metrics
- CPU usage
- Memory usage
- Pod status

Step 10: Verify Prometheus

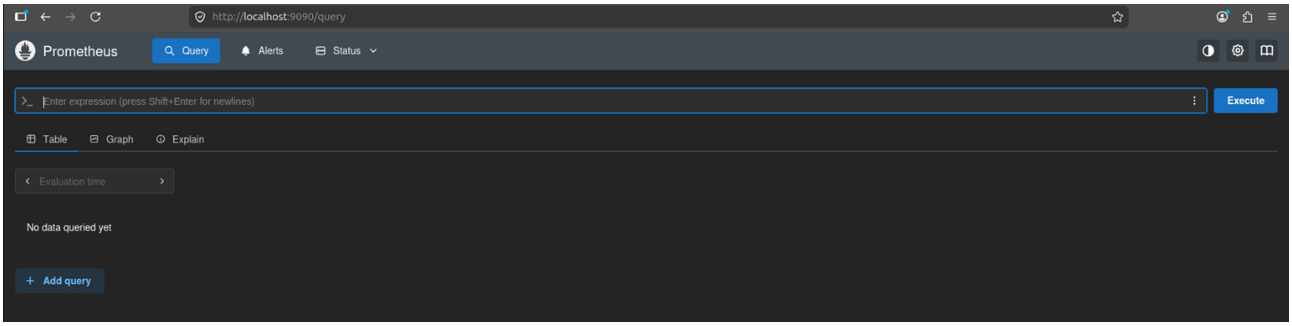
Port forward:

```
kubectl port-forward svc/monitoring-kube-prometheus-prometheus
9090
```



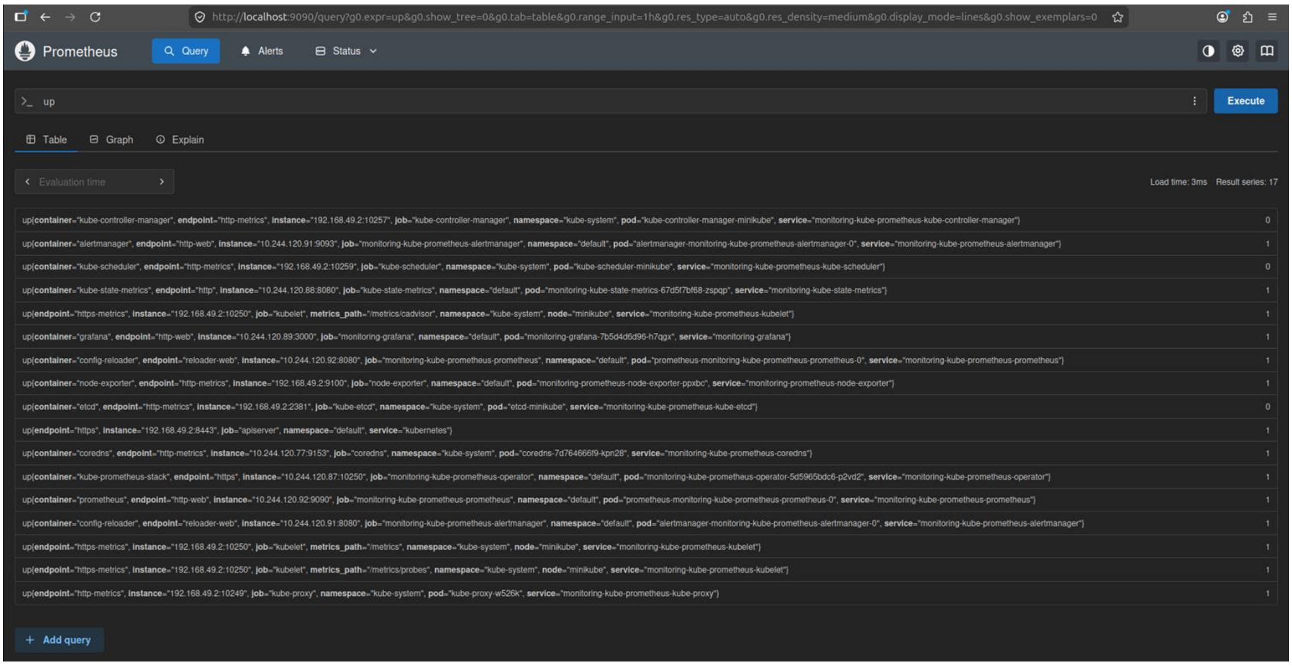
Open:

`http://localhost:9090`



Try query:

`up`



Expected Output:

- Grafana dashboard loads
- Metrics visible (CPU, memory, pods)
- Prometheus queries working